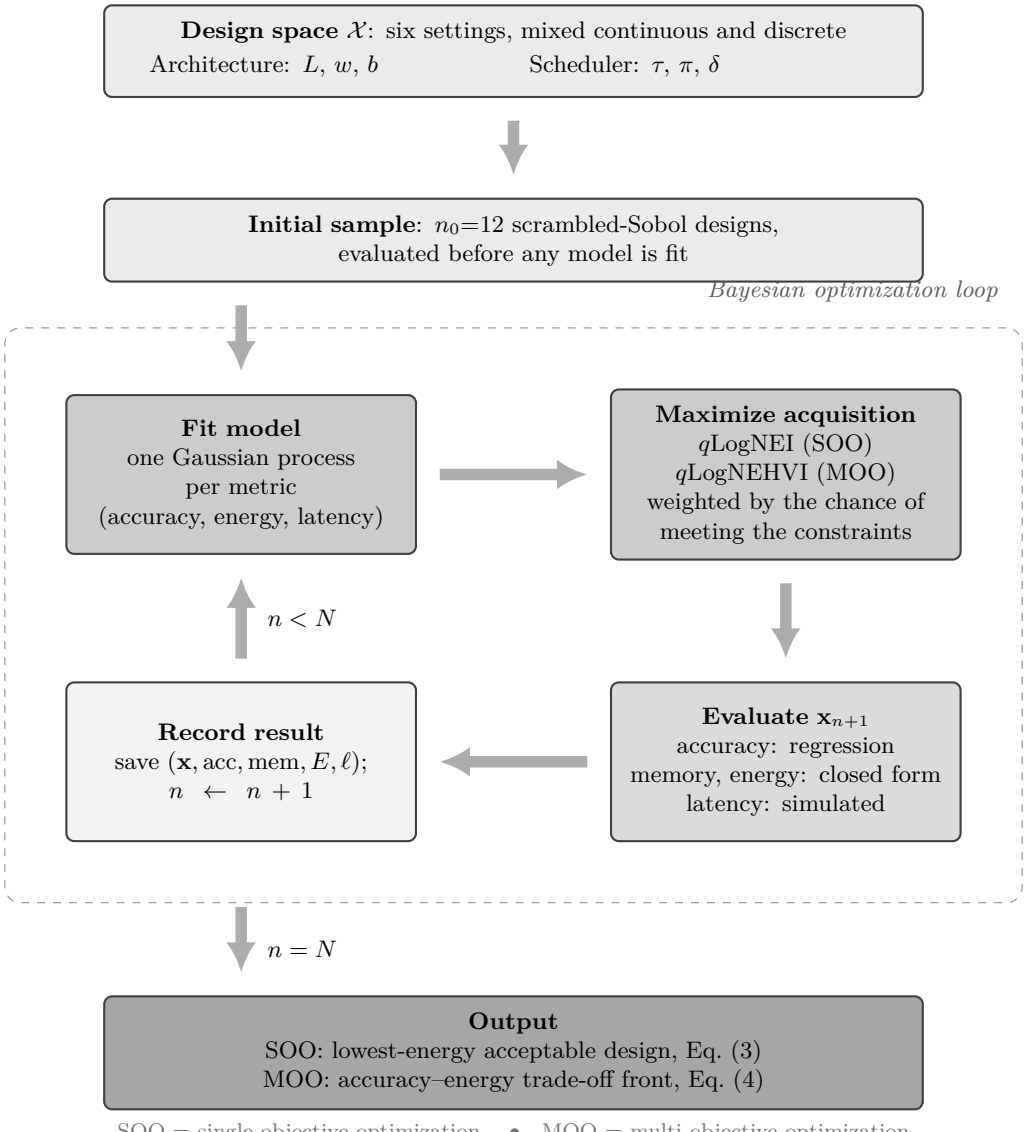
Design space $\mathcal{X}$: six settings, mixed continuous and discrete
Architecture: L, w, b
Scheduler: τ, π, δ
Initial sample: n_0=12 scrambled-Sobol designs, evaluated before any model is fit
Bayesian optimization loop
Fit model
one Gaussian process per metric (accuracy, energy, latency)
Maximize acquisition
qLogNEI (SOO)
qLogNEHVI (MOO)
weighted by the chance of meeting the constraints
$n < N$
Record result
save $(\mathbf{x}, \text{acc}, \text{mem}, E, \ell)$;
$n \leftarrow n + 1$
Evaluate $\mathbf{x}_{n+1}$
accuracy: regression
memory, energy: closed form
latency: simulated
$n = N$
Output
SOO: lowest-energy acceptable design, Eq. (3)
MOO: accuracy–energy trade-off front, Eq. (4)
SOO = single-objective optimization • MOO = multi-objective optimization